

Inverse of 3x3 matrices

Key Points

To find the inverse of a 3x3 matrix use

$$M^{-1} = \frac{1}{|M|} \text{Adj}(M)$$

Where $|M|$ is the determinant and $\text{Adj}(M) = C^T$ is the transposed cofactor matrix C made of elements alternating in sign multiplied by the determinant of the minors of M

$$c_{ij} = (-1)^{i+j} |M_{i,j}|$$

Basic Skills

Q1. If possible find the inverse of these 2x2 matrices

a) $A = \begin{pmatrix} 4 & -2 \\ 1 & 7 \end{pmatrix}$ **b)** $B = \begin{pmatrix} 6 & 8 \\ 3 & 4 \end{pmatrix}$

Q2. What condition makes a matrix M invertible

Q3. For the matrix $M = \begin{pmatrix} 6 & 5 & -3 \\ 3 & 0 & 2 \\ -2 & 1 & 10 \end{pmatrix}$

- a)** Write down the minors $M_{1,3}$ and $M_{2,2}$
- b)** Calculate the cofactors c_{13} and c_{22}
- c)** Calculate $|M|$
- d)** Decide if the matrix M is invertible or not

Competency Skills

Q4. For the matrix $M = \begin{pmatrix} 3 & 0 & 1 \\ 2 & 0 & -2 \\ 0 & 5 & 1 \end{pmatrix}$

- a)** Work out $|M|$
- b)** Work out all the minors determinants $|M_{i,j}|$ (for $i, j = 1, 2, 3$) and hence form the cofactor matrix C
- c)** Write down the $\text{Adj}(M)$ and hence find M^{-1}
- d)** Check your answer by using $MM^{-1} = I$

Q5. If possible find the inverse of the following matrices

a) $A = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 5 & -6 \end{pmatrix}$ **b)** $B = \begin{pmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{pmatrix}$ **c)** $C = \begin{pmatrix} 1 & 5 & 3 \\ 2 & 4 & -2 \\ 4 & 2 & 1 \end{pmatrix}$

Advanced Skills

Q6. Given the following matrices $A = \begin{pmatrix} 1 & -2 & k \\ 2 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$, $B = \begin{pmatrix} -5 & -2+2k & -4-k \\ 8 & -1-3k & -2+2k \\ 1 & -8 & 5 \end{pmatrix}$

- a)** State the values of k such that it is possible to find A^{-1}
- b)** For any valid value of k by calculating AB , or otherwise, find A^{-1}
- c)** Hence solve the matrix equation $Av = \begin{pmatrix} 1 \\ 12 \\ 3 \end{pmatrix}$
- d)** Using $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ write down the equivalent set of 3 simultaneous equations you solved in part c)

Extension

Q7. By using the same approach try to work out the inverse of a 4x4 matrix of your choice

Hint: The formula/method still holds

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